Chapter : 4 Discrete Filter Structure

$* \; h[n] = 0 \quad \text{for } n \geq M \leftarrow (\text{Def}^n\text{: FIR})$

Causal FIR:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] \; \{\text{original LTI system}\}$$

$$= \sum_{k=0}^{M-1} h[k]x[n-k]$$

$$= \sum_{k=0}^{M-1} b_k x[n-k] \text{ where, } x = \text{multiplier of } b_k$$

* Causal FIR Filter Realization

$$y[n] = \sum_{k=0}^{M-1} b_k x[n-k]$$

$$\Longrightarrow \frac{Y(Z)}{X(Z)} = H(Z) = \sum_{k=0}^{M-1} b_k Z^{-k}$$

(2.) Cascade - Form:

$$H(Z) = \sum_{k=0}^{M-1} b_k Z^{-k}$$

$$\text{Modify: } \; H(Z) = \prod_{k=1}^{\kappa} H_k(Z)$$

where,

$$H_k(Z) = b_{k_0} + b_{K_1} Z^{-1} + b_{k_2} Z^{-2} \; \& \; \text{k} = \text{integer part of } \; \frac{M+1}{2}$$

(3.) Lattice Structure:

$$H(Z) = \text{given}$$

$$H_m(Z) = A_m(Z) \quad , \quad m = \text{order of FIR Filter}$$

$$A_m(Z) = A_{m-1}(Z) + K_m Z^{-1} B_{m-1}(Z)$$

$$B_m(Z) = K_m A_{m-1}(Z) + Z^{-1} B_{m-1}(Z)$$

$$\text{Solve: } \; A_{m-1}(Z) = \frac{A_m(Z) - K_m B_m(Z)}{1 - (K_m)^2}$$

$$B_m(Z) = Z^{-m} A_m(Z^{-1})$$

$K_m =$ Lattice Coefficient $=$ $\alpha_m(m)$ coefficient / multiplier of Z^{-m} in $A_m(Z)$.

$*$ if $K_1, K_2, ..., K_n$ given ,

$$A_0(Z) = B_0(Z) = 1$$

$$A_m(Z) = A_{m-1}(Z) + K_m Z^{-1} B_{m-1}(Z)$$

$$B_m(Z) = K_m A_{m-1}(Z) + Z^{-1} B_{m-1}(Z)$$

IIR Filter Structure:

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

$$H(Z) = \frac{Y(Z)}{X(Z)} = \frac{\sum_{k=0}^{M} b_k Z^{-k}}{1 + \sum_{k=1}^{N} a_k Z^{-k}}$$

$$= H_1(Z).H_2(Z)$$

$$\text{where, } H_1(Z) = \sum_{k=0}^{M} b_k Z^{-k}$$

$$H_2(Z) = \frac{1}{1 + \sum_{k=1}^{N} a_k Z^{-k}}$$

Cascade Structure

$$H(Z) = \prod_{k=1}^{P} H_k(P)$$

where

$$H_k(P) = \frac{b_{k_0} + b_{k_1} Z^{-1} + b_{k_2} Z^{-2}}{1 + a_{k_1} Z^{-1} + a_{k_2} Z^{-2}}$$

$$\& \text{ P} = \text{integer part of } \left(\frac{N+1}{2}\right)$$

(3.) Lattice or Lattice-Ladder Structure:

$$H(Z) = \frac{P(Z)}{Q(Z)}$$

$$\text{Let } A_m(Z) = Q(Z) \Longrightarrow \text{Do as in FIR}$$

$$C_{m-1}(Z) = C_m(Z) - U_m B_m(Z) \quad , \quad U_m = C_m(m)$$

$$\text{where } C_m(Z) = P(Z).$$

$$U_m = \text{Ladder Coefficient}$$

Chapter : 5

FIR Filter Design

Common Window:

1. Rectangular :

$$w(n) = \begin{cases} 1 \ , \ 0 \leq n \leq M \\ 0 \ , \ \text{otherwise} \end{cases}$$

2. Bartlett :

$$w(n) = \begin{cases} \frac{2n}{M} & , \ 0 \leq n \leq \frac{M}{2} \\ 2 - \frac{2n}{M} & , \ \frac{M}{2} < n \leq M \\ 0 & , \ \text{otherwise} \end{cases}$$

3. Hanning :

$$w(n) = \begin{cases} 0.5 - 0.5\cos\left(\frac{2\pi n}{M}\right) \ , \ 0 \leq n \leq M \\ 0 \ , \ \text{otherwise} \end{cases}$$

4. Hamming :

$$w(n) = \begin{cases} 0.54 - 0.46\cos\left(\frac{2\pi n}{M}\right) \ , \ 0 \leq n \leq M \\ 0 \ , \ \text{otherwise} \end{cases}$$

5. Blackman :

$$w(n) = \begin{cases} 0.42 - 0.5\cos\left(\frac{2\pi n}{M}\right) + 0.08\cos\left(\frac{4\pi n}{M}\right) \ , \ 0 \leq n \leq M \\ 0 \ , \ \text{otherwise} \end{cases}$$

Main Lobe Width	Peak Approx. Error or stopband attenuation (dB)
$\frac{4\pi}{(M+1)}$	-21
$\frac{8\pi}{M}$	-25
$\frac{8\pi}{M}$	-44
$\frac{8\pi}{M}$	-53
$\frac{12\pi}{M}$	-74

Linear - Phase Pass FIR Filter:

Desired freq. response:

$$H_{\text{lp}}(\text{jw}) = H_{\text{lp}}(e^{\text{jw}})$$

$$= \begin{cases} e^{\text{jw}\frac{M}{2}} \text{ , } |w| \le w_c \\ 0 \quad\quad \text{ , } w_c < |w| \le \pi \end{cases}$$

$$h_{\text{lp}} = F^{-1}\left[H_{\text{lp}}(\text{jw})\right]$$

$$= \frac{\sin\left[w_c\left(n-\frac{M}{2}\right)\right]}{\pi\left(n-\frac{M}{2}\right)}$$

$$= \frac{w_c}{\pi}\frac{\sin\left[w_c\left(n-\frac{M}{2}\right)\right]}{w_c\left(n-\frac{M}{2}\right)}$$

$$= \frac{w_c}{\pi}.\,\text{sinc}\left[w_c\left(n-\frac{M}{2}\right)\right].$$

symmetric: $h_{\text{lp}}[n] = h_{\text{lp}}[M-N]$

Kaiser Window:

$$w[n] = \begin{cases} I_0 \frac{\left[\left(\beta\sqrt{1-\left[\left(\frac{n-\alpha}{\alpha}\right)^2\right]}\right)\right]}{I_0(\beta)} & , 0 \le n \le M \\ 0 & , \text{otherwise} \end{cases}$$

$$\alpha = \frac{M}{2}$$

$I_0[.]$ = Zeroth Order Modified Bessel Function of First Kind.

$$I_0[x] = 1 + \frac{0.25x^2}{(1!)^2} + \frac{\left(0.25x^2\right)^2}{(2!)^2} + \frac{\left(0.25x^2\right)^3}{(3!)^2} + ...$$

$$\Delta \text{w} = w_2 - w_1 = w_s - w_p$$

$$A = -20\log_{10}(\delta) \ , \ \delta = \min(\delta_1, \delta_2)$$

$$\beta = \begin{cases} 0.1102(A - 8.7) & , A > 50 \\ 0.5842(A-21)^{0.4} + 0.07886(A-21) & , 21 \le A \le 50 \\ 0 & , A < 20. \end{cases}$$

$$M = \frac{A - 8}{2.285\Delta \text{w}}$$

$$w_c = \frac{w_p + w_s}{2}$$

$$= \frac{w_1 + w_2}{2}$$

stopband attenuation $(A_s) = -20\log_{10}(\delta), \delta = \min(\delta_1, \delta_2)$

Transition Width $\Delta \text{w} = w_2 - w_1 = w_s - w_p$

Δw = Main lobe width.

$$\Delta \text{w} = \frac{8\pi}{M} \Longrightarrow \text{Ex.}$$

$$w_c = \frac{w_p + w_s}{2}$$

N = filter length

M = filter order $= N - 1$

For Hamming:

$$\boxed{\Delta = w_s - w_c}$$

//.

Chapter : 6

Design of IIR Filter

(i) IIR Filter Design using Impulse Invariance.

Analog Filter: $H_a(s) = \sum_{k=1}^{N} \frac{C_k}{S - P_k}$, C_k = constants , P_k = Poles .

inverse Laplace Transform

$$h_a(t) = \sum_{k=1}^{N} C_k . e^{P_k . t}$$

$$h[n] = \sum_{k=1}^{N} C_k . e^{P_k \eta T}$$

Z-Transform

$$H(Z) = \sum_{n=-\infty}^{\infty} h[n] . Z^{-n}$$

$$= \sum_{n=0}^{\infty} h[n] Z^{-n}$$

$$= \sum_{k=1}^{N} \frac{C_k}{1 - e^{P_k T} . Z^{-1}}$$

Replace $S - P_k$ by $1 - e^{P_k T} Z^{-1}$

//.

(2.) Bilinear Transformation

$$\text{Let} : H_a(s) = \frac{b}{s + a}$$

.
.
.

$$H(Z) = \frac{b}{\left(\frac{2}{T}\right)\left(\frac{1 - Z^{-1}}{1 + Z^{-1}}\right) + a}$$

$$\therefore s = \frac{2}{T} \frac{1 - Z^{-1}}{1 + Z^{-1}}$$

$$= j\Omega$$

$$\Rightarrow \Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right).$$

$$\Rightarrow \omega = 2 \tan^{-1}\left(\frac{\Omega T}{2}\right).$$

Some commonly used Analog Filters & Their Characteristics:

(1.) Butterworth (LPF) Filter:

simple, maximally-flat filter.

$$* \, |H(\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

Magnitude Response.

$$= \frac{1}{1 + \varepsilon^2 \left(\frac{\Omega}{\Omega_p}\right)^{2N}}$$

$$\Omega_c = \text{cut-off freq} = \text{3dB freq.}$$

$$N = \text{Order of filter}$$

$$\varepsilon = \text{Scaling factor } (\leq 1)$$

$$* \text{ Poles} : S_k = \Omega_c e^{\frac{j\pi}{2}} . e^{\frac{j(2K+1)\pi}{2N}} \ , \ K = 0, 1, 2, ..., N-1$$

$$* \text{ Order} : N = \frac{\frac{1}{2}\log\left(\frac{1}{(1-\rho_p)^2} - 1\right) - \frac{1}{2}\log\left(\frac{1}{\delta_s^2} - 1\right)}{\log\left(\frac{\Omega_p}{\Omega_s}\right)}$$

not in dB. $\begin{cases} \rho_p = \delta_p = \delta_1 \\ \rho_s = \delta_s = \delta_2 \end{cases}$

OR:

$$N = \frac{\frac{1}{2}\log\left[\frac{10^{0.1A_s} - 1}{10^{0.1A_p} - 1}\right]}{\log\left(\frac{\Omega_s}{\Omega_p}\right)}$$

$$A_p = \text{Passband ripple / attenuation}$$

$$A_s = \text{Stopband ripple / attenuation}$$

$$\Omega_c = \frac{\Omega_p}{\left[\frac{1}{(1-\rho_p)^2} - 1\right]^{\frac{1}{2N}}}$$

$$H(s) = H_a(s) = \frac{\Omega_c^N}{(s - s_0)(s - s_1)...}$$

Chebyshev (LPF) Filter:

2 types:→

(i) Chebyshev = Chebyshev - Type I

(ii) Inverse Chebyshev = Chebyshev - Type II

Type - I

$$* \, |H(\Omega)|^2 = \frac{1}{1 + \varepsilon^2 T_N^2 \left(\frac{\Omega}{\Omega_p} \right)}$$

where,

$$.T_N(x) = \begin{cases} \cos(N \cos^{-1} x) & , \, |x| \leq 1 \\ \cosh(N \cosh^{-1} x) & , \, |x| > 1 \end{cases})$$

$$.\varepsilon = \left[\frac{1}{\left(1 - \rho_p\right)^2} - 1 \right]^{\frac{1}{2}}$$

$$\text{Poles} : x_k + j y_k \ , \ k = 0 \text{ to } (N-1)$$

where,

$$x_k = r_2 \cos \varphi_k$$

$$y_k = r_2 \sin \varphi_k$$

where,

$$r_2 = \Omega_p . \frac{\beta^2 - 1}{2\beta^2} \ , \ \beta = \left(\frac{\sqrt{1 + \varepsilon^2} + 1}{\varepsilon} \right)^{\frac{1}{N}}$$

$$\varphi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N} \ // \ .$$